TIC 390780000

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        1.617150 d
  Epoch (t0)    1654.7497 BTJD
  Duration      0.5000 d
  Depth         3.199 %  (31,988 ppm)
  SNR           167.8

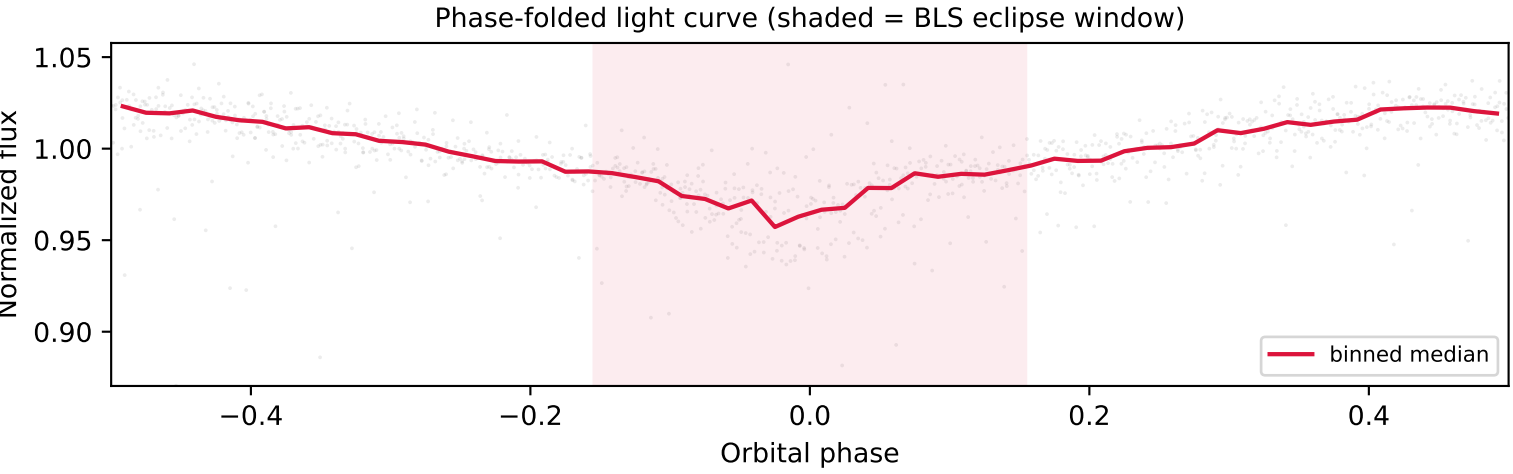
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      18  (gate >= 3)
  Robust depth ratio     0.93  (gate >= 0.50)
  Robust depth sigma     29.4  (gate >= 7.0)
  Odd/even z-score       2.18  (consistent)
  Secondary eclipse      not detected
```

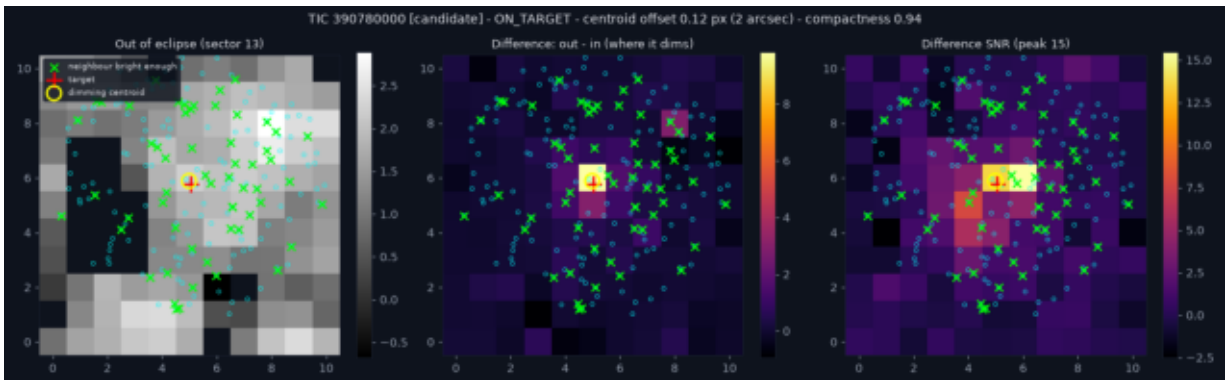
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 13    light-curve dip in this sector: 28.9 sigma    t0 refined by -0.0150 phase
  Difference-image peak SNR   15.47   (gate >= 5.0)      Compactness   0.94   (gate >= 0.50)
  Dimming centroid offset     0.12 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 390779995
    0.62 px away, dTmag +1.66 - farther from the dimming than the target
```





No caveats - clean detection on every check.